



Pyrolysis of waste animal fats in a fixed-bed reactor: Production and characterization of bio-oil and bio-char



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ABSTRACT

Several animal (lamb, poultry and swine) fatty wastes were pyrolyzed under nitrogen, in a laboratory scale fixed-bed reactor and the main products (liquid bio-oil, solid bio-char and syngas) were obtained. The purpose of this study is to produce and characterize bio-oil and bio-char obtained from pyrolysis of animal fatty wastes. The maximum production of bio-oil was achieved at a pyrolysis temperature of 500 °C and a heating rate of 5 °C/min. The chemical (GC–MS analyses) and spectroscopic analyses (FTIR analyses) of bio-oil showed that it is a complex mixture consisting of different classes of organic compounds, i.e., hydrocarbons (alkanes, alkenes, cyclic compounds...etc.), carboxylic acids, aldehydes, ketones, esters,...etc. According to fuel properties, produced bio-oils showed good properties, suitable for its use as an engine fuel or as a potential source for synthetic fuels and chemical feedstock. Obtained bio-chars had low carbon content and high ash content which make them unattractive for as renewable source energy.

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1. Introduction

Renewable energy sources are being considered as an attractive solution for overcoming the energy needs and for reducing the environmental problems. Compared to other renewable energy, biomass is the most interesting energy source since it can be converted directly into liquid, gaseous and solid fuels, usable for transport, heat and power production (Bridgwater and Peacocke, 2000; Asadullah et al., 2007; Blin et al., 2007; Goyal et al., 2008). Among a large variety of biomass resources, animal fatty wastes (AFW) are precious materials for second generation bio-fuels production as they are formed mainly of triglycerides, with high energetic value (Maher and Bressler, 2007).

In Tunisia, the amount of AFW generated, in 2010, was about 80.000 tons (ANGed, 2010). Regarding the method of treatment or disposal, a small percentage of AFW is recycled (around 22%) while its major part is buried in sanitary landfills, which are associated with environmental problems (leachate and biogas production). These environmental problems can be alleviated by converting these AFW into energy.

A wide range of technologies such as biochemical, thermochemical, and physical and/or chemical processes are available for recovering bio-fuels from triglyceride based materials. Thermochemical processes include gasification, pyrolysis and combustion

(Bridgwater, 2003; Goyal et al., 2008). Among these technologies, pyrolysis is thought to be more favorable for converting AFW into mainly valuable liquid hydrocarbons (Maher and Bressler, 2007; Wisniewski et al., 2010). Pyrolysis is a thermal decomposition of organic substances under oxygen-deficient circumstances into various phases: liquid products (condensable vapors at cooling temperature); carbon-rich solid residues (bio-char); gaseous products (syngas which were not condensable gases) (Maschio et al., 1992; Bridgwater, 2003). The liquid product (bio-oil or pyrolytic oil), which is a complex mixture of oxygenated hydrocarbons and water, can be used directly as a liquid fuel or as source of synthetic chemical feedstocks (Bridgwater, 2003; Blin et al., 2007; Maher and Bressler, 2007). Pyrolysis bio-oil from fish fatty wastes has already been successfully tested as a direct fuel in Diesel engines (Varuvel et al., 2012). The solid bio-char, similar to fossil coal, is also a useful product that can be used as bio-fuel (high calorific value) or chemical adsorbent (as a substitute for activated carbon) (Maschio et al., 1992; Bae et al., 2011) or as soil amendment (Grierson et al., 2011). Moreover, the pyrolysis products quality and distributions depend mainly on some experimental parameters: pyrolysis final temperature, heating rate, residence time, type of pyrolysis reactor, type of raw material...etc. (Zanzi et al., 1996; Bridgwater, 2003). The influence of the process parameters and the pyrolysis products properties have been widely discussed for various biomass resources (Asadullah et al., 2007) but, few literature is available for fatty materials pyrolysis products (yields and properties). Maher and Bressler (2007) in a review on the

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thermo-chemical conversion of triglyceride materials, presented the state of the art of different pyrolysis processes, pyrolytic products properties and applications, and concluded that the obtained bio-oils composition are entirely different as a function of the feedstock. Wisniewski et al. (2010), comparing bio-oils produced from triglycerides materials and lignocellulosic biomass, showed that the first one contained mainly of alkanes, alkenes, ketones, aldehydes, aromatics and carboxylic acids and the second one was formed basically of phenols, benzenediols, furanes and their derivatives.

However, compared to lingo-cellulosic materials, the pyrolysis of fatty wastes is not very established in the literature. Main research developed in the area of bio-fuels from fatty materials concerns bio-diesel production through transesterification process (Srivastava and Prasad, 2000; Fukuda et al., 2001; Demirbas, 2003; Tashtoush et al., 2004; Phan and Phan, 2008; Sabudak and Yildiz, 2010). This study aimed (i) to investigate the effects of main parameters (temperature and heating rate) and also of the nature of feedstock, on the pyrolysis products distribution; (ii) to determine the suitable conditions for the production of the maximum of bio-oil; and (iii) to characterize bio-oils and bio-chars obtained from several AFW pyrolysis in the suitable conditions.

2. Materials and methods

2.1. Raw material

The animal (lamb, poultry and swine) fatty wastes, used in this study for pyrolysis feed, were obtained from various meat factories around Tunis, in northern of Tunisia. Prior to pyrolysis experiments, animal wastes were freeze-dried and analyzed using elemental analysis (CHNS-O). The CHN-contents of the input materials were determined via Perkin Elmer 2400 CHN elemental analyzer. The sulfur content was measured using Horiba Jobin Yvon elemental sulfur analyzer. The oxygen content was calculated by difference.

The fatty acids composition of raw material was determined using a Hewlett–Packard (HP 6890) GC with FID detector. It is equipped with a capillary column type HP innowax (30 m × 0.25 mm × 0.25 µm). The carrier gas was Helium. The column temperature was programmed from 40 to 300 °C at 5 °C/min after an initial 1 min isothermal period and kept at the final temperature for 30 min. The inlet was set at 250 °C, and sample injection was made in split mode (1:60). The device is controlled by a HP BRIO operator “HP Chemstation” for varying the physical parameters. Before GC analysis, fatty acids methyl esters (FAME) solutions were prepared from several wastes by means of “cold methylation” method. Ash contents of raw materials were measured using calcination method according to NF T 60-111 standard protocol.

2.2. Pyrolysis experiments

The pyrolysis experiments were carried out in a laboratory scale fixed bed reactor heated by an electric furnace. The reactor was made of stainless steel with 30 cm height and 15 cm internal diameter. The temperature was controlled by a thermocouple immersed inside the reactor. All pyrolysis experiments were carried out under atmospheric pressure and inert conditions. During experiments, the reactor is continuously flushed with nitrogen in order to remove air from the reactor and all the gases produced during pyrolysis.

In all experiments, approximately 200 g of animal fats, placed in batch in the reactor, were heated from room temperature until the desired temperature at different heating rates and held at that desired temperature until no further significant release of gas

was observed (approximately for 20 min). The pyrolysis experiments were repeated three times and the products yields presented are the mean value of three equivalent experiments.

Gas and vapors formed during pyrolysis, flow out from the top of the reactor, and then passed through a trapping system. The trapping system consists of two condensers cooled with the circulation of glycoled water, in order to reduce the temperature to around −5 °C. The liquid products from the two condensers were collected in conical flasks fitted to the condensers. The liquid fraction (bio-oil) consisted of two phases: aqueous (bottom phase) and organic rich compounds (upper phase) which were separated by decantation. The yield of collected bio-oil was calculated by “The amount of bio-oil collected/total amount of initial feedstock X 100”.

The remaining bio-char was collected after completing the pyrolysis reaction when the room temperature is reached in the reactor. The bio-char yield was calculated by “The amount of bio-char collected/total amount of initial feedstock X 100”.

The non-condensable vapors passed through a cleaning system formed by a water bubble in order to condensate remaining hydrocarbons and water. The purified gas mixture passed through a filtering system consisting of a column of activated carbon and gas filters in order to retain particles. After particles, tars and water removal, the gases were forwarded for analysis using a gas analyzer equipped with (i) an infrared detector for CO, CO₂, CH₄, C_nH_m analyses, (ii) a thermal conductivity detector to quantify H₂ and (iii) an electrochemical cell for O₂ measurement. The amount of non-condensable gases produced was determined by difference from mass balance.

2.3. Bio-oil and bio-char characterization

2.3.1. Elemental analysis and ash content

The elemental composition (CHNS-O) and ash content of the bio-chars were determined with the same methods used for raw materials.

2.3.2. Fourier Transform InfraRed (FTIR) spectroscopy

FTIR spectroscopic analyses were performed to determine the distribution of functional groups present in pyrolysis products (bio-oils and bio-chars). The FTIR spectra of produced bio-oils and bio-chars were recorded in the absorbance mode, between 4000 and 400 cm^{−1}, on pellets made from a mixture of Bio-oil/KBr, using a Perkin Elmer Spectrum BX spectrophotometer.

Several organic groups show characteristic absorption peaks originating from vibrational transitions in the molecules. These absorption bands are characteristic for the class of compound (e.g. aliphatics, aromatics, alcohols, acids, esters, amides,...etc.) and can be used for both qualitative and quantitative analysis of complex samples containing a mix of organic compounds. The identification of these absorptions peaks was made according the previously published data.

2.3.3. Fuel properties

The bio-oil fuel properties (acid value, pH, calorific value, viscosity, density, water content and ash content) were measured in the upper organic fraction of bio-oil using the similar standard procedures used for petroleum products.

Acid value indicates the acidity level of bio-oils. It was measured according to the acido-base titration method (ISO 6618). The pH was measured with a digital pH meter according to pH-Potentiometer method. The Higher Heating Value (HHV) of bio-oil was measured using an oxygen bomb calorimeter according to NF M 07-030 standard method. The viscosity of bio-oil was measured using a petroleum products kinematic viscosity analyzer (a Cannon–Fenske Routine Viscometer). The density of the bio-oil

was analyzed using hydrometer method (ISO 3675). The water content in the bio-oil was measured by distillation method (NF T 60-113). The ash content was measured by burning the bio-oil sample in a Muffle furnace according to the NF T 60-111.

2.3.4. Bio-oil fractionation

The complex mixture of bio-oil was separated into a neutral, an acidic and a polar fraction by SPE using AminoPropyl Bond Elute® cartridges. Each fraction was eluted successively with a mixture (1:1, v:v) of Dichloromethane-Methanol for neutral fraction, a mixture (9:1, v:v) of Ether-Formic acid for acidic compounds and only Methanol for polar compounds. Each fraction was dried and weighed and then stored in the freezer for molecular analyses.

2.3.5. Mass spectrometry-gas chromatography (GC-MS)

GC-MS analyses of the neutral fraction from bio-oils were performed using an Agilent 7890 Model gas chromatograph equipped with Agilent 5975 mass selective detector. The gas chromatograph was fitted with a HP-5 MS capillary column (30 m × 0.25 mm i.d., 0.25 µm film thickness). The injector was set at 250 °C and helium was the carrier gas. The GC oven was hold at 40 °C for 2 min then heated to 300 °C at a rate of 5 °C min⁻¹ and finally 30 min hold at 300 °C. The identification of the compounds was accomplished by advising a NIST database and by comparing published mass spectra.

2.3.6. X-RAY diffraction method

The crystal structures of bio-chars obtained with different animal fatty wastes were examined by a diffractometer (PANanalytical X'Pert PRO).

3. Results and discussion

3.1. Feed materials characteristics

The main characteristics of studied animal fat wastes are summarized in Table 1. All studied materials have high carbon contents (up to ~75% in the three studied samples), moderate oxygen contents (mean value of 19.3% for the three samples) and high hydrogen contents (around 12% for the three studied samples) but low nitrogen and sulfur contents (avg. close to 0.58% and 0.19%, respectively). The ash content of the input material are also very low (not exceed 0.64%).

The fatty acids compositions of studied animal fat wastes, given in Table 1, show that major fatty acids are Octadecenoic acid (C18:1) > Hexadecanoic acid (C16:0) > Octadecanoic acid (C18:0). The predominance of the C18:1 and C16:0 fatty acids was also highlighted by numerous studies focused on the waste animal fats

chemical characterization (Sreenivasan, 1968; Rice et al., 1997; Varuvel et al., 2012).

3.2. Products yields

3.2.1. Effect of type of raw material

The pyrolysis of several animal fat wastes produced a variation in products yields. Table 2 shows relative products yields (compared to the initial sample mass (wt.%)) from the pyrolysis of AFW of different origin (lamb, poultry and swine). The effect of raw material types can be seen from the comparison of the bio-oil yields. As can be seen in Table 2, the pyrolysis of lamb fats produced the highest liquid fraction yield of 77.9 wt.% while poultry and swine wastes gave lower amounts of condensable fraction, around 67.6 and 58.0 wt.%, respectively (Table 2).

The liquid fraction consisted of two phases: aqueous (bottom phase) and organic-rich (upper phase) which were separated by decantation. For the three studied samples, the organic-rich phase had the highest percentage (between 45.8 and 61.6 wt.%) compared to the bottom one which presented little amounts (between 12.2 and 21.5 wt.%) as shown in Table 2.

These results are in agreement with previous studies on pyrolysis of different feedstocks: woody biomass (bio-oil yields between 60 and 75 wt.%, Mohan et al., 2006); marine algae (bio-oil yields between 37.5 and 47.4 wt.%, Bae et al., 2011); poly-ethylene based materials (bio-oil yields around 86.2 wt.%, Grieco and Baldi, 2012); sewage sludge (bio-oil yields between 27 and 54 wt.%, Fonts et al., 2012).

Higher yields of bio-oil were recorded by other authors working with similar type of wastes (triglycerides based materials): Demirbas (2007) reported 77.1 wt.% of bio-oil for pyrolysis of beef tallow at 500 °C; Wiggers et al. (2009) obtained a bio-oil yield around 73 wt.% for fish oil wastes at 525 °C; Wisniewski et al. (2010) reported 72–73 wt.% of bio-oil from waste fish oil pyrolysis.

The water content of the produced liquid fraction (between 12.2 and 21.5 wt.%, Table 2), is in the range of those published by many authors (water content between 15 and 30 wt.%) for pyrolytic oils from different biomass resources (Bridgwater, 2003; Onay and Kocchar, 2003; Demirbas, 2007; Ndiaye, 2008).

The water contained in the pyrolytic oils was derived from the moisture of the raw material (dehydration reaction while thermal cracking). This water content has an influence on the fuel properties of bio-oils and namely on their heating value and thus on their use as engine fuel (ignition and injection problems) (Demirbas, 2007). In order to minimize the water content in the liquid fraction, it is recommended that the moisture content of feedstock be reduced (raw material drying).

3.2.2. Effect of final temperature and heating rate on yields of pyrolysis products

Two different sets of pyrolysis experiments were carried out in order to investigate the effects of both parameters: the temperature of pyrolysis and the heating rate on pyrolysis products distributions. The first set was performed, for each type of animal fats, at different final temperatures (400, 450, 500 and 550 °C) and at a constant heating rate of 5 °C/min. The second set were conducted

Table 1
Characteristics of the animal fatty wastes.

	Lamb	Poultry	Swine
<i>Elemental composition (%)</i>			
C	74.63	63.25	65.38
H	12.11	11.26	11.32
O	12.50	23.64	21.99
N	0.15	1.03	0.58
S	0.27	0.23	0.09
Ash content (%)	0.34	0.59	0.64
<i>GC-MS composition</i>			
Palmitic acid (C16:0)	28%	22%	16%
Stearic acid (C18:0)	23%	–	11%
Oleic acid (C18:1)	32%	34%	18%
Linoleic (C18:2)	–	15%	–
Behenic acid (C22:0)	–	–	10%
Other acids	17%	29%	45%

Table 2
Pyrolysis products yields (wt.%) for different animal fatty wastes.

	Liquid fraction		Solid fraction	Gaseous fraction
	Organic phase	Aqueous phase		
Lamb	61.6	16.3	1.8	20.3
Poultry	46.1	21.5	1.1	31.3
Swine	45.8	12.2	1.5	40.5

at different heating rates (15 °C/min, 10 °C/min, 8 °C/min and 5 °C/min) while the final temperature was maintained at 500 °C under nitrogen atmosphere. The results of relative yields of pyrolysis products from the two experiments series, for lamb fatty wastes, are summarized in Figs. 1 and 2.

At low temperature (400 °C) the fraction of bio-oil was 54 wt.% and the fraction of bio-char was 32 wt.% and the syngas fraction was 14 wt.%. By increasing the pyrolysis temperature, the fraction of bio-oil gradually increased while the fraction of bio-char decreased; the gas fraction also showed a slight increase (from 14 wt.% at 400 °C to 21 wt.% at 550 °C) as shown in Fig. 1. Moreover, no significant effect on the bio-oil yields (around 76 wt.%) of the increase of pyrolysis temperature between 500 °C and 550 °C, was observed (Fig. 1).

Similar effects of the final temperature on the pyrolysis products yields were highlighted by numerous authors (Maschio et al., 1992; Horne and Williams, 1996; Ozbay et al., 2001; Asadullah et al., 2007). In these pyrolysis studies, using various types of organic wastes and different experiment sets, the increase of final temperature is always followed by an important increase in the bio-oil yield, a decrease of bio-char yield and a slight increase in the gases amounts. For example, Asadullah et al. (2007), studied the pyrolysis of bagasse and evaluated the effect of final temperature on product distribution: the increase of pyrolysis temperature from 300 °C to 600 °C, induced an increase in the bio-oil yield (from 18.66 wt.% to 66.63 wt.%) and a decrease of bio-char yield (from 77 wt.% to 24.66 wt.%), gaseous products also increased (from 4.34 wt.% to 14.71 wt.%).

The effects of the heating rate on the pyrolysis products distribution are almost the opposite of those observed for pyrolysis tem-

perature: with the increasing of the heating rate (from 5 °C/min to 15 °C/min), the bio-oil yield decreased significantly (from 80 wt.% to 5 wt.%) while the bio-char yield increased (from 3 wt.% to 33 wt.%) and the syngas proportion increased (from 17 wt.% to 62 wt.%) (Fig. 2). These behaviors are due to the liquid phase cracking due to the increasing of the heating rate. In fact, the long carbon chains contained in condensable vapors can be broken into short ones, inducing an increase in gaseous products. For heating rate, different trends were reported by authors working with different type of wastes: Inguanzo et al. (2002), studied the pyrolysis of sewage sludge and established that the higher the pyrolysis heating rate, the more efficient the pyrolysis (a higher production of pyrolytic oil). Moreover, Mohan et al. (2006), reported for woody materials that the increase of the heating rate induce an enhancement of bio-oil production.

The above comparison suggests that the pyrolytic behavior resulting from the heating rate variation can be associated to the raw feedstock composition: fatty materials differ from lignocellulosic biomass in their constitutional compounds (mainly fatty acids), high C, O, H and low ash contents. Thus, in the case of triglycerides materials, a high heating rate can induce a cracking of condensable vapors.

As mentioned in the introduction, the purpose of the pyrolysis experiments is to maximize the production of liquid fraction. Therefore, all these experiments indicate that the pyrolysis temperature of 500 °C and the rate of 5 °C/min are the most convenient conditions to get better results (maximum bio-oil yield around 62%) from the pyrolysis of several AFW.

For further analysis, only bio-oils and bio-chars samples that were produced in the most convenient experimental conditions (final temperature pyrolysis about 500 °C and heating rate of 5 °C/min) were retained for next characterization.

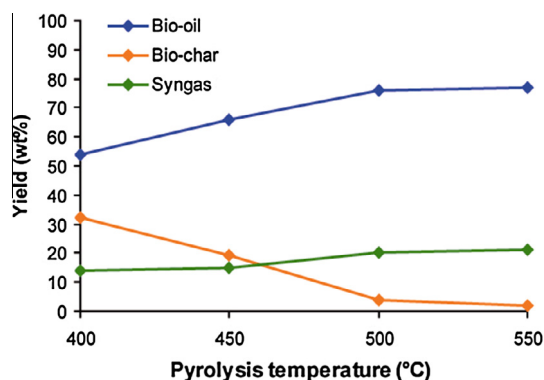


Fig. 1. The effect of final temperature on pyrolysis products distribution (lamb fatty wastes, invariable heating rate 5 °C/min).

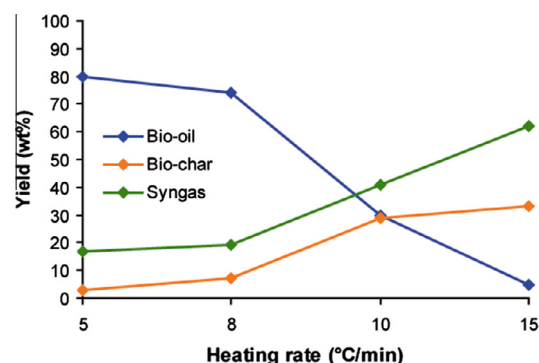


Fig. 2. The effect of heating rate on pyrolysis products distribution (lamb fatty wastes, invariable final temperature of 500 °C).

3.3. Bio-oil characterization

3.3.1. FTIR characterization

The obtained FTIR spectra for all studied bio-oils show some common features in absorptions bands but with a little difference in the relative intensity. As an example, FTIR spectrum obtained from lamb fatty wastes bio-oil is given in Fig. 3. The presented spectrum exhibits several absorption bands that can be attributed to many functional groups. The O–H stretching vibration bands between 3200 and 3600 cm^{-1} indicate the presence of carboxylic acids and alcohols as well as the presence of water (Fig. 3).

The presence of intense bands around 2920 cm^{-1} and 2850 cm^{-1} and the two shoulders around 2960 cm^{-1} and 2870 cm^{-1} , attributable to symmetric and asymmetric stretching of C–H in CH_2 and CH_3 groups, respectively, are well distinguished in all the bio-oil samples. The high intensity of these bands is a marker of the dominance of saturated aliphatic hydrocarbons in the studied bio-oils. The presence of other aliphatic groups (C–H bending vibration bands at 1384 cm^{-1} and asymmetric deformation of C–H between 1475 and 1430 cm^{-1}), with moderate intensity, confirm the aliphatic character of the studied bio-oils. The presence of bands at 1620–1680 cm^{-1} , which are attributed to C=C stretching vibrations, is an indicator of the presence of aryl substituted (unsaturated) alkenes (Grierson et al., 2011). The C=O stretching vibration bands in the frequency range of 1770–1700 cm^{-1} is present in all the spectra and indicate the presence of ketones, carboxylic acid, aldehydes or esters. The bands at 950 and 1300 cm^{-1} are assigned to C–O stretching vibrations, indicating the presence of alcohol or esters. Several signals appear in the range of 700 and 900 cm^{-1} , attributable to the aromatic stretching vibrations, indicating the presence of aromatic compounds in bio-oils.

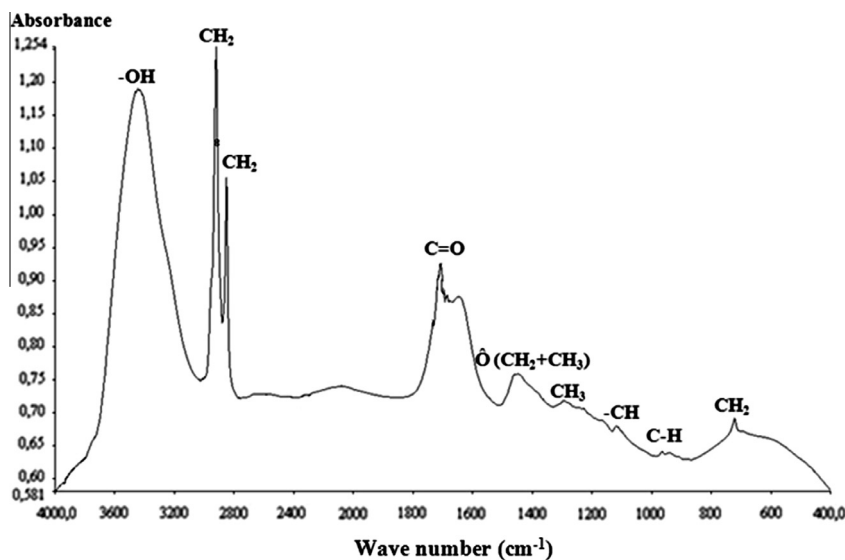


Fig. 3. FTIR spectra of bio-oil obtained from lamb fatty wastes.

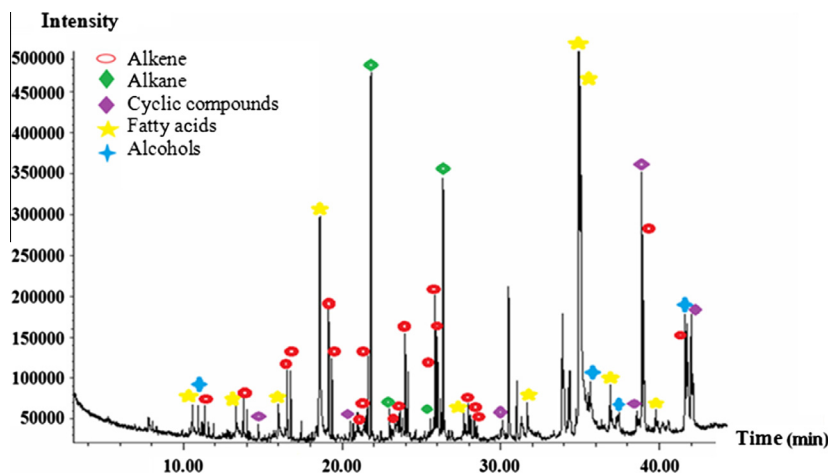


Fig. 4. GC-MS chromatogram of neutral fraction of bio-oil obtained from swine fats pyrolysis.

The obtained FTIR spectra are similar to other bio-oils FTIR spectra already described in published work (Rutkowski and Kubicz, 2006; Kalisz et al., 2008; Zhou et al., 2010; Grierson et al., 2011; Lu et al., 2011; Malumbaz et al., 2011) for various biomass resources pyrolysis. From various bio-oils FTIR results, these liquid products exhibit a wide variety of chemical groups (aliphatic saturated and unsaturated compounds; oxygenated groups).

The FTIR results show the complexity of the bio-oil chemical composition and the multiplicity of organic compounds constituting this liquid product. The application of FTIR absorption spectroscopy is very helpful and yet quite simple, to have a rapid characterization of bio-oils composition.

3.3.2. Bio-oil composition

The yields of neutral, acidic and polar fractions of the studied bio-oil are presented in Table 3. As it can be seen in Table 3, all studied bio-oils are composed mostly of neutral fraction (up to ~98 wt.%) which is composed mainly of hydrocarbons (linear, branched and cyclic hydrocarbons) of ketones, aldehydes, alcohols, esters, sterols... etc.

In order to have the detailed molecular characterization, further analysis using GC-MS were conducted on the major neutral fraction. The GC-MS profile of the neutral fraction of the bio-oil ob-

Table 3

Bio-oils fractions (neutral, acidic and polar).

	Neutral fraction (%)	Acid fraction (%)	Polar fraction (%)
Lamb	95.9	1.8	2.3
Poultry	97.9	0.6	1.5
Swine	87.3	11.9	0.8

tained from swine fatty wastes is illustrated in Fig. 4. The major compounds detected were numbered (listed according to retention time increasing) and their area determined (Table 4).

The identification of major components of the neutral fraction of bio-oil products shows the presence predominantly of these compounds families: carboxylic acids (Heptanoic acid, Octanoic acid, Nonanoic acid, Decanoic acid... etc.); alkanes (Dodecane, Tridecane, Tetradecane, Pentadecane, Hexadecane... etc.); alkenes (1-Dodecene, 1-Tridecene, 2-Tetradecene, 6-Tridecene, ... etc.) and cyclic hydrocarbons (1-Cyclodecene, Bicyclopentyl... etc.). Many other compounds were identified but in little amounts (ketones, aldehydes, alcohols... etc.).

As shown in Table 4, for studied sample, the carbon distribution of alkanes is ranged from C11 to C18 and from C9 to C22 for alkenes.

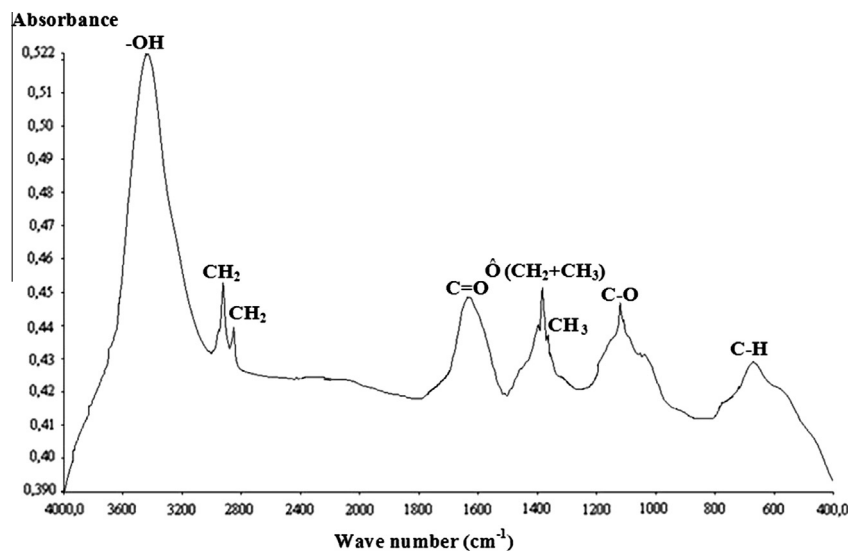


Fig. 5. FTIR spectra of bio-char obtained from swine fatty wastes.

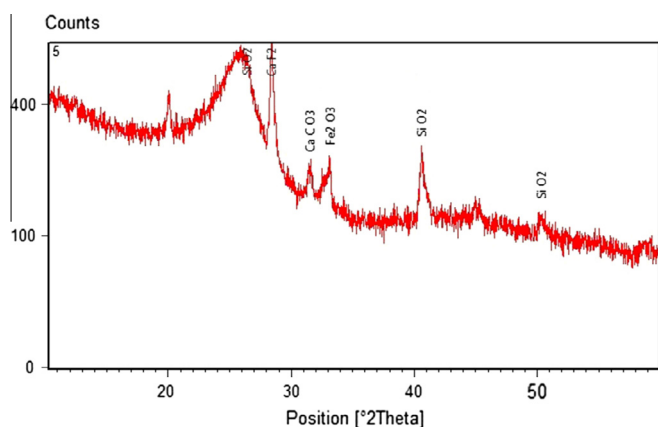


Fig. 6. DRX-diagram of bio-char obtained from swine fatty wastes.

In comparison to GC–MS compositions of bio-oils produced from triglycerides materials pyrolysis, great similarities in the detected compounds can be found (Srivastava and Prasad, 2000; Lima et al., 2004). For example, in a review of fuel compositions from triglyceride materials pyrolysis, Srivastava and Prasad (2000) reported that fatty materials pyrolysis produces several compounds including alkanes, alkenes, alkadienes, aromatics and carboxylic acids. Lima et al. (2004), studying the chemical composition of pyrolytic products from soybean and castor oils, established that hydrocarbons and oxygenated organic compounds such as alkanes, alkenes, alkadienes, and carboxylic acids are the predominant compounds in the studied samples.

For bio-oils from lignocellulosic biomass, the molecular composition is quite different since typical chemical substances related to the original polymers – cellulose, hemicellulose, and lignin – can be found (such as phenolic compounds: phenols, guaiacols, syringols and furans) besides the fatty acids, alkanes, alkenes, amides, aldehydes (Mohan et al., 2006).

The mechanism of the thermal decomposition of triglycerides has been largely discussed by Maschio et al. (1992), Srivastava and Prasad (2000) and Maher and Bressler (2007), who recognized the great complexity of pyrolysis reactions and the occurrence of some secondary reactions. In the case of fatty materials pyrolysis, the most important chemical reaction that leads to the formation

of alkanes and alkenes is the cleavage of the carboxyl side of triglycerides entities, followed by the decarboxylation, however, the occurrence of carboxylic acids is usually associated to the cleavage of the glycerol moieties (Maher and Bressler, 2007). The formation of aromatic moieties is generally attributed to secondary reactions (cyclization of olefin structures and Diels–Alder reactions).

The obtained results show that the neutral fraction of pyrolytic oil exhibits a wide variety of chemical substances (carboxylic acids, alkanes, alkenes, aldehydes, ketones, alcohols), which is in good agreement with FTIR results.

3.3.3. Fuel properties

The fuel properties (acid value, pH, Higher Heating Value, viscosity, density, water content and ash content) were determined for the organic upper phase of bio-oil samples obtained under most convenient conditions (500 °C and 5 °C/min) from swine and poultry fatty wastes. The obtained fuel properties are listed in Table 5. For comparison purposes, the specifications of Tunisian Diesel (according to the JORT standard) and European biodiesel (according to the EN14214 standard) were also presented in Table 5, since the pyrolytic oil from triglyceride materials is likely to approach Diesel fuels.

As observed in Table 5, the acid values of the organic phase of studied bio-oils are higher (124.34 mg KOH/g of sample for poultry wastes and 138.95 mg KOH/g of sample for swine wastes) than the required values (for example, for Tunisian Diesel, the acid value must not exceed 5 mg KOH/g of sample; according to the EN14214 standard for bio-fuels, the acid value for biodiesel must not exceed 0.5 mg KOH/g). Similar high acid values (from 107 to 142 mg KOH/g of sample) were reported by Wiggers et al. (2009) for different fractions of bio-oil obtained from waste fish oil.

The pH value of studied bio-oils is close to 3, indicating the acid character of these products. These pH values are very low compared to petroleum fuel (pH is required to be near neutrality). For bio-oils produced from woody materials, very low pH values (between 2.4 and 2.8, Oasmaa and Czernik, 1999 and around 2.5, Mohan et al. 2006) were also obtained.

In general, the acid value of bio-oils indicates the presence of free fatty acids in the studied sample as it was observed with FTIR and GC–MS analyses. In order to be useful as engine fuel (to avoid corrosion of combustion boilers and turbines), the content of acidic components in the bio-oil should be decreased. Catalytic pyrolysis

Table 4

Compounds identified in the neutral fraction of bio-oil produced from swine fatty wastes.

No.	Name of the compound	Area%	No.	Name of the compound	Area%
1	Heptanoic acid	0.54	29	Tetradecanoic acid	0.69
2	1-Undecanol	0.40	30	(9E)-9-Octadecene	0.88
3	2-Undecene	0.49	31	9-Octadecene	0.43
4	Octanoic acid	0.75	32	1-Octadecene	0.45
5	1-Dodecene	0.72	33	1,6-Cyclodecadiene	0.50
6	Dodecane	0.53	34	Hexadecanenitrile	3.62
7	1-Cyclododecene	0.34	35	Methyl Palmitate	1.20
8	Nonanoic acid	0.87	36	Cyclopentadecanone	0.58
9	1-Tridecene	1.00	37	n-Hexadecanoic acid	0.93
10	Tridecane	1.10	38	Oleanitrile	2.89
11	Decanoic acid	7.66	39	Methyl 11-Octadecenoate	0.77
12	2-Tetradecene	2.06	40	Octadecanenitrile	1.78
13	Tetradecane	1.31	41	Heptadecanoic Acid	0.61
14	Bicyclopentyl	0.36	42	6-Octadecanoic Acid	15.54
15	1H-Indene	0.40	43	9-Octadecanoic Acid	8.31
16	6-Tridecene	0.36	44	1-Pentadecene	0.34
17	1-Pentadecene	1.36	45	1-Nonodecan-1-ol	0.53
18	Pentadecane	6.12	46	11-Hydroxyundecanoic acid	0.78
19	n-Nonylcyclohexane	0.66	47	1-Eicosanol	0.34
20	Cyclohexadecene	0.78	48	1-Cyclohexylheptene	0.45
21	(3Z)-3-Hexadecene	0.57	49	Cycloteracosane	5.16
22	1-Hexadecene	1.84	50	1-Docosene	3.17
23	Hexadecane	1.20	51	9-Octanoic acid	0.48
24	1-Heptadecene	0.33	52	1,13-Tetradecadienne	2.87
25	8-Heptadecene	2.59	53	11-Hexacosyne	1.61
26	3-Heptadecene	2.23	54	Henicosan-1-ol	2.25
27	1-Heptadecene	0.71	55	1,4-Didecylcyclohexane	1.03
28	Heptadecane	4.53			

Table 5

Fuel properties of bio-oils produced from of poultry and swine fatty wastes.

	Unit	Bio-oil		Tunisian Diesel (JORT, 2007)	European Biodiesel (EN 14214)
		Poultry	Swine		
Acidity index	mg KOH/g of sample	124.34	138.95	5	0.5
pH	–	2.9	2.2	Neutrality	–
Calorific value	kcal/kg	9638	9516	11,000–15,000	
Viscosity at 40 °C	cSt	5.230	5.648	2–4.5	3.5–5
Density at 15 °C	kg/l	0.8862	–	0.81–0.89	0.86–0.90
Flash point	°C	46	40	55–120	>101
Water content	%	0.7	0.9	Traces	0.5
Ash content	%	0	0	0.01	0.02

is one of solution for decreasing acidity in bio-oils, and thus enhancement of bio-oil properties.

The Higher heating value (HHV) of the organic phase of studied bio-oils is around 9600 kcal/kg, slightly lower than those of petroleum fuel (for Tunisian Diesel: calorific value range from 11,000 to 15,000 kcal/kg). According to [Oasmaa and Czernik \(1999\)](#), bio-oils heating values are about 40–45% of that of petroleum fuel oil. Furthermore, the HHV of studied samples compares with the previous findings in relation of fatty materials: [Varuvel et al. \(2012\)](#) reported calorific value around 9314 kcal/kg for bio-oils produced from fish fatty wastes. The obtained HHV in this study are slightly higher than those obtained for bio-oils from various biomass resources. For example, [Bae et al. \(2011\)](#), reported slightly lower HHV (between 5565 and 8025 kcal/kg) for the organic upper fraction of bio-oil obtained from macroalgae. In the [Grierson et al. \(2011\)](#) study, the HHV of bio-oils samples produced from microalgae was around 6687 kcal/kg. Moreover, [Mohan et al., 2006](#), in a review on properties of bio-oils produced from woody biomass, given more low heating values (between 3822 and 4538 kcal/kg) for these products.

For the studied samples, the bio-oil calorific value is not very high and this may be due to its high oxygen and moisture content.

The viscosity at 40 °C values of the organic phase of studied bio-oils, around 5 cSt, are slightly higher than those of petroleum

Diesel (between 2 and 4.5 cSt, for standard norms of Tunisian Diesel and between 3.5 and 5 for standard norms of European Diesel) but they are in the range values of viscosity of bio-oils from fatty materials: [Wiggers et al. \(2009\)](#) reported a viscosity value of 4.8 cSt at 40 °C for heavy fraction of bio-oil from waste fish oil; [Srivastava and Prasad \(2000\)](#) studied pyrolytic oil from soya bean oil pyrolysis and reported a viscosity value around 10.2 cSt at 38 °C. Moreover, for bio-oils produced from woody materials, [Oasmaa and Czernik \(1999\)](#) reported higher viscosity values (from 13 to 28 cSt at 50 °C). Viscosity, since it is the measurement of the liquid internal friction, is an important parameter that must be considered for the use of bio-oils as engine fuel.

The density of studied samples, around 0.89 kg/l, is found to be within the range of the densities of petroleum products (between 0.81 and 0.89 kg/l for Tunisian Diesel and between 0.86 and 0.90 kg/l for European Diesel), indicating that this bio-oil is useful directly as engine fuel. This density value is also comparable of those published by [Wiggers et al. \(2009\)](#) and [Varuvel et al. \(2012\)](#) who separately obtained density values about 0.87 kg/l and 0.89 kg/l for bio-oils from pyrolysis of fish wastes.

Flash point values of studied samples are between 40 and 46 °C. These values are almost lower than those of Tunisian Diesel between 55 and 120 °C and for European Diesel >101 °C ([Table 5](#)). [Oasmaa and Czernik \(1999\)](#) reported for bio-oils produced from

woody materials, higher flash point values comprising between 62 and 95 °C.

The water content of the upper organic phase of bio-oil is around 0.08% whereas the ash content is very low (near zero). The bio-oil water content is directly related to raw material moisture content.

The fuel properties of the upper organic phase of studied bio-oils are quite similar to that of Petroleum Diesel. The HHV and the density of the organic-rich fraction of bio-oils from fatty wastes pyrolysis suggest that these products are useful directly as fuel and have the potential to be used as a renewable energy resource. However, compared with other bio-fuels properties – biodiesel namely-, it was found that the studied samples had some undesired properties especially for their handling and their application as fuel: high acid value, high viscosity, high water and ash contents. Therefore, further improvement of these fuel properties before its application as an engine fuel, is required. Catalytic pyrolysis is proposed for upgrading these bio-oil properties.

3.4. Bio-char characterization

3.4.1. Elemental composition and ash content

All studied materials have moderate carbon content (15.67%), high oxygen content (57.33%), low hydrogen (3.47%) and nitrogen (2.18%) contents and very low sulfur (0.32%) contents. The ash content of the studied bio-chars is high (around 21.03%). The bio-char low carbon content and high ash content make it inconvenient for use as renewable fuel.

3.4.2. FTIR characterization

An example of FTIR spectrum obtained for bio-char sample produced from swine fatty wastes pyrolysis, in suitable conditions, is given in Fig. 5. The obtained bio-char spectrum shows a markedly aliphatic character, indicated by the presence of absorption bands between 3000 and 2800 cm^{-1} , a slight peaks at 1470 cm^{-1} and 1380 cm^{-1} , imputable, respectively, to asymmetric and symmetric vibrations of C-H groups. The aromatic nature of bio-chars is indicated by the presence of the absorption band at 1628 cm^{-1} imputable to C=C groups vibrations and a series of little peaks in the 700–900 cm^{-1} range, attributable the aromatic stretching vibrations. Finally, the presence of O-containing groups is revealed by: (i) the O–H stretching vibration bands between 3200 and 3600 cm^{-1} ; (ii) the C=O at 1636 cm^{-1} and (iii) the bands around 1120 cm^{-1} attributable to C–O stretching.

In the studied bio-char spectrum, large peaks recorded at low frequency below 600 cm^{-1} are attributable to various inorganic compounds remaining in the ash.

All these obtained bands in bio-char spectrum can be found in the raw feedstock, which consists namely of triglycerides with high amounts of C, O and H.

3.4.3. XRD analysis

In this study, the X-ray diffraction method was applied in order to analyze the bio-chars crystallinity. XRD pattern of bio-char sample obtained from swine fatty wastes is shown in Fig. 6. This XRD analyses showed that bio-char sample contains many minerals (or inorganic) species, such as Quartz (SiO_2), Hematite (Fe_2O_3), Calcite (CaCO_3), Fluorite (CaF_2), Sodium Chloride (NaCl).

4. Conclusion

This study confirms that animal fatty wastes are precious raw materials for bio-oil production by pyrolysis process. The investigation of the effects of experimental parameters on the quality and the distribution of pyrolysis products indicate that the

temperature of 500 °C and the heating rate of 5 °C/min are the most convenient conditions to get better results (maximum bio-oil yield) from the pyrolysis of animal fats wastes. In these suitable conditions, the yields of bio-oils obtained from studied animal fatty wastes range from 58 wt.% to 77.9 wt.%. Bio-oils FTIR characterization reveals: (i) high aliphatic character (presence of straight chain hydrocarbons), (ii) high oxygenated groups content and (iii) low aromatic nature of the studied bio-oil indicating that these liquid products can be used as engine fuel (aliphatic hydrocarbons required for the application of the bio-oil as engine fuel). The characterization of bio-oil fractions obtained in the best conditions, by GC–MS analysis indicate that studied bio-oils are a complex mixture with a large variety of organic components (alkanes, alkenes, cyclic hydrocarbons, carboxylic acids, aldehydes, ketones, alcohols, esters, ... etc.) indicating that they can be used in chemical industry (chemical feedstocks). The fuel properties of the upper organic phase of studied bio-oils, quite similar to that of Petroleum Diesel, confirm that these products are useful as fuel and have the potential to be used as a renewable energy resource. But prior to their application, these oils should be upgraded to improve physical properties. Catalytic thermal cracking is also suggested for decreasing bio-oils acidity and increasing their heating value. Bio-chars FTIR characterization and elemental analyses reveal mainly a high oxygenated character but a moderate aliphatic character making their utilization as solid fuel (coke) not very attractive.

It can be concluded that pyrolysis is a successful method of converting fatty wastes and giving a high yield mainly of liquid products (hydrocarbons). However, the obtained bio-oils still require enhancement in composition and fuel properties before their application as engine fuels, but they can be precious sources of synthesis of chemical products.

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